

Mechanistic Interpretability of ASR models using Sparse Autoencoders

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Abstract

Understanding the internal machinations of deep Transformer-based NLP models is more crucial than ever as these models see widespread use in various domains that affect the public at large, such as industry, academia, finance, health. While these models have advanced rapidly, their internal mechanisms remain largely a mystery. Techniques such as Sparse Autoencoders (SAE) have emerged to understand these mechanisms by projecting dense representations into a sparse vector. While existing research has demonstrated the viability of the SAE in interpreting text-based Large Language Models (LLMs), there are no equivalent studies that demonstrate the application of a SAE to audio processing models like Automatic Speech Recognizers (ASRs). In this work, a SAE is applied to Whisper, a Transformer-based ASR, training a high-dimensional sparse latent space on frame-level embeddings extracted from the Whisper encoder. Our work uncovers diverse monosemantic features across linguistic and non-linguistic boundaries, and demonstrates cross-lingual feature steering. This work establishes the viability of a SAE model and demonstrates that Whisper encodes a rich amount of linguistic information.

1 Introduction

In the last few years, advances in the Transformer architecture have led to models with the ability to produce text, audio, image and video. These impressive capabilities come with an increased need to understand their internal representations. While there have been many studies focused on understanding how text is generated by an LLM, for example by monitoring the chain-of-thought reasoning tokens, it has remained challenging to interpret the internal representations of these models.

Despite this difficulty, there have been some approaches that have seen success. For example, one form of mechanistic interpretability that has shown promise is the Sparse Autoencoder (SAE), which uses a sparse latent space to examine semantically meaningful information (Bricken et al., 2023b). SAE models have been successful in analyzing the type of information present in densely encoded data produced by intermediate layers of neural network models. Recent work has focused largely on large language models, however the application of this technique can be used on a wide variety of different types of models to extract interpretable information from previously uninterpretable machine output.

While SAEs have been successfully applied to text-based LLMs, it is unclear whether they can successfully be applied to interpret audio-based models, such as Whisper. Whisper, an automatic speech recognition model (ASR), is capable of transcribing and translating spoken word in a wide variety of languages (Radford et al., 2022). While it would be expected that only linguistic information is encoded by Whisper, there has been evidence that other information such as noise is used to condition the decoding of the transcript (Gong et al., 2023). Whisper possibly encodes a wide variety of information that is not

necessarily required in order to transcribe speech, but may facilitate transcription. The use of SAEs may enable better understanding of the types of information that are being encoded by the audio models.

In this work we explore the use of a SAE model to probe the internal data representations of an ASR model, specifically Whisper. The results suggest that the SAE captures a range of distinct representational patterns, providing evidence for the diversity of representations encoded by Whisper. The key contributions of this work are threefold:

1. To the best of our knowledge, we believe this is the first work that applies a SAE-based mechanistic interpretability technique to an ASR model, with the goal of understanding features across linguistic and non-linguistic boundaries, and enable feature steering.
2. Our results show that Whisper learns a rich and diverse set of representations from audio input, including phonetic, semantic, morphological, positional, and lexical features.
3. Our findings demonstrate the utility of SAE as an effective interpretability tool for probing the internal representations of Whisper.

2 Related Work

There has been a substantial body of research attempting to understand the internal representations of large language models (Tenney et al., 2019; Haley, 2020; Li & Wisniewski, 2021; Lasri et al., 2022; Yao et al., 2025; Misra & Mahowald, 2024; Houghton, 2025; Pan & Bergen, 2025; Houghton et al., 2025). While these results have generated valuable insights into the internal representations of large language models, they have often relied on examining the probability distribution of output tokens in carefully designed studies.

Recently, however, researchers have found success with mechanistic interpretation methods for inspecting the internal workings of models. One such approach is the use of sparse autoencoders (SAEs) to decompose the densely encoded information of a model into sparse monosemantic features. Much of this work has focused on interpretability in text-based Large Language Models (LLMs) as a means to better understand what information is being encoded in these large and complex models (Bricken et al., 2023a; Huben et al., 2024; Templeton et al., 2024; Gao et al., 2025). For example, Bricken et al. (2023a) demonstrated the effectiveness of training a Sparse Autoencoder on the internal activations of a 1-layer transformer model. They found that a Sparse Autoencoder trained this way learns monosemantic features such as text written in Arabic. Further, Huben et al. (2024) built upon this and demonstrated the effectiveness of Sparse Autoencoders on even more complex language models. Specifically, they demonstrated that a Sparse Autoencoder trained on the internal activations of Pythia (Biderman et al., 2023) learns human-interpretable features, such as a feature for court cases and a feature for last names. Sun et al. (2025) found that a SAE is capable of learning intrinsically dense features of an LLM, while Chanin et al. (2025) examined SAE feature splitting and absorptions.

While these studies demonstrate that SAEs are a useful tool for interpreting the internal representations of text-based LLMs, the evidence beyond this is mixed. It remains unclear, for instance, whether this approach extends to other modalities, such as end-to-end ASR models that take audio rather than text as their input. Some prior work has probed the internal representations of Whisper by building a classifier from each layer of the encoder (Gong et al., 2023), and others have applied the Sparse Autoencoder to a speaker recognition model (Pluth et al., 2025). Neither of these, however, investigated linguistic representations within an audio processing model. There is therefore no convincing demonstration to date that the SAE is an effective tool for interpreting ASR models. Our work addresses this gap.

3 Method

In order to examine the internal representations of Whisper, a SAE was applied immediately after the final layer of the encoder. Since the final layer of the encoder is directly processed by the decoder, it potentially contains the most important information in order for the decoder to accurately transcribe the audio.¹ Figure 1 shows the Whisper architecture with the SAE inserted. This analysis focuses entirely on the latent space of the SAE, with the Whisper decoder only being utilized to examine the effects on the transcript after modifying the latent values.

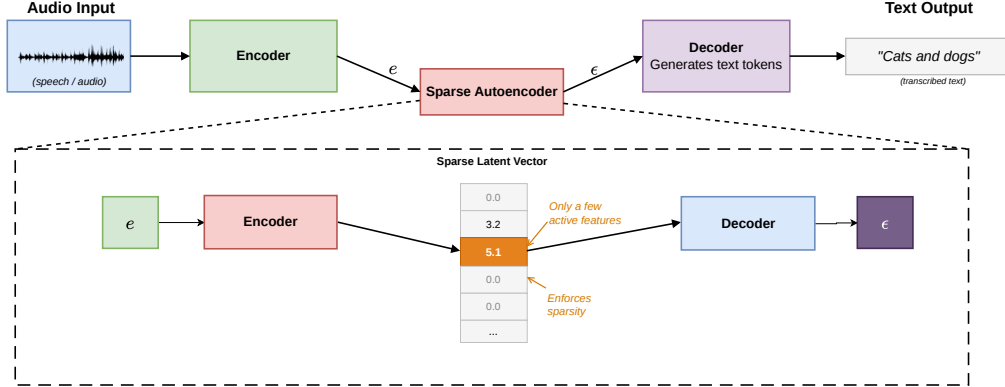


Figure 1: Diagram of the Whisper architecture. The SAE is trained based on the outputs of the Whisper encoder. When steering is performed, the reconstructed embedding can be used as an input to the Whisper decoder to generate manipulated text.

The SAE training framework used was developed based on the OpenAI sparse autoencoder² which implements a k-Sparse Autoencoder (Makhzani & Frey, 2014; Gao et al., 2025). The SAE model encoder is a simple linear layer with a topK activation. The SAE decoder is another linear layer which reconstructs the Whisper embedding space. The SAE effectively maps the 512 width Whisper base encoding to a large hidden layer of 16000 elements and then back to the 512 elements. In order to constrain sparsity, the model was trained using a TopK activation to constrain the number of latents which can be activated, in this work $K=45$. This creates a total of 16.4 million trainable parameters³. The SAE is trained using mean squared error (MSE) as the loss function between the input and output of the SAE.⁴

Since Whisper takes an input of 30 seconds of audio, all audio files are padded or trimmed to 30 seconds by default. To avoid training the SAE on padded frames, they are identified and removed during SAE training. The final input data to the SAE is specifically the individual frame-level Whisper embedding cut to the audio length. A given audio will generate up to 1500 frames (30 seconds worth) of training data.

3.1 Dataset

The data used to train the SAE comprised several different datasets, since it is expected that data diversity is critical to help develop a diverse set of latent features. This dataset comprises LJSpeech (Ito & Johnson, 2017), LibriSpeech (Panayotov et al., 2015), Voxceleb 1 (Nagrani et al., 2017), Mozilla Common Voice English (Ardila et al., 2020), SLR39 (Morgan, 2006), SLR67 (Hernandez-Mena, 2019), SLR61 and SLR71-SLR75 (Guevara-Rukoz et al., 2020), and Musan (Snyder et al., 2015). In total the dataset comprises 646,769 files. Since the

¹Although in principle, the SAE could be situated between any layers in the encoder or decoder.

²<https://github.com/openai/sparse-autoencoder>

³2 layers with $512 * 16000$ parameters each

⁴The training code, analysis code, analysis dataset and latent examples with automated labels can be found in this anonymous repo: <https://anonymous.4open.science/r/COLM2026-73CF/>

data points are the individual time frames from the Whisper encoder, there are on the order of 200 million data points which is sufficiently many to train the approximately 16 million parameters of the SAE.

Additional data was used in some analyses to evaluate the trained model, these datasets include Mozilla Common Voice Spanish and French.

3.2 Steering

Steering is a causal analysis that demonstrates that modifying a given latent (or latents) in the latent space has a specific and predictable effect on downstream tasks. For example, in [Figure 1](#), the sparse latent vector would be modified then passed to the SAE decoder to produce ϵ' . This modified encoding can then be taken as an input to the Whisper decoder to produce a manipulated transcript. This is a powerful technique to demonstrate that the latent index is a representation of the hypothesis feature, since the causal relationship is clearly visible through differences between the transcript from before steering the latent and after steering the latent.

In order to identify features, the latent space was explored for correlations between latents and various linguistic and non-linguistic features, including language features, phonetic features, lexical features, syntactic features, morphological features, and semantic features. A latent can be considered to correspond to a given feature if its activation is predictive of that feature. For each hypothesized feature, precision and recall values were computed to quantify what proportion of activated files contain the hypothesized feature. For some features, causality was confirmed through feature steering, which demonstrates that a latent was not simply correlated with a given feature, but directly encode that feature.

3.2.1 Analysis Set

To support this analysis, a dedicated set of 28,414 audio files was constructed. The analysis set comprises 13,084 files from LJSpeech-1.1, 7,830 English and 7,500 Spanish files randomly sampled from Mozilla Common Voice (v13.0), and 331 additional clips from a curated profanity subset drawn from Mozilla Common Voice English. [Table 1](#) summarizes the composition. Although the test set is a subset of the SAE training data, this is inconsequential. The SAE is trained solely to reconstruct Whisper embeddings, meaning any structure it learns is already present in Whisper’s representations. Evaluating the SAE on data that it has already seen does not change the kinds of information present in the Whisper embeddings. A formal explanation of this claim is provided in [Appendix A](#).

Dataset	Source	N
LJSpeech	LJSpeech-1.1	13,084
Common Voice (English)	CV Corpus 13.0	7,830
Common Voice (English, profanity)	CV Corpus 13.0	331
Common Voice (Spanish)	CV Corpus 13.0	7,500
Total		28,414

Table 1: Analysis set composition.

In the case of the phoneme, morpheme, and lexical analyses, the dataset was subset to use the LJSpeech corpus. LJSpeech has TextGrids that align the audio to phone-level transcriptions, which enables the examination of phonological sequences that may be spelt differently (e.g., the /aɪ/ sound in *light* vs *sky*).

For all other analyses, we used the entirety of the test set. In those cases, alignment was performed using a forced-alignment tool to produce character- and word-level alignments for each file. This was done, for those analyses, for the LJSpeech corpus as well in those analyses for the sake of consistency.

3.2.2 Automated Labeling

To supplement the manual feature analysis, all latents were labeled using GPT-OSS 120B. For each latent, the 100 files closest to the median activation strength were selected as representative examples, and up to 20 were presented to the model with activated character spans marked by asterisks (e.g., `grad*ual*ly`). The model was prompted to assign one of seven categories: phonetic, orthographic, morphological, lexical, semantic, syntactic, or diffuse for the examples. The model subsequently returns a structured JSON response containing a label, explanation, confidence level, and category. Only high-confidence labels were retained; lower-confidence responses were assigned the *diffuse* label with the leading hypothesis preserved. A random subset of 100 of these latents was manually evaluated to determine the quality of the automated labeling. The automatically labeled latents yielded an accuracy of 76%. These results are reported in [Appendix B](#).

4 Whisper SAE Results

The results of the feature analyses are presented in this section. The SAE reveals that Whisper encodes a surprisingly rich hierarchy of representations, spanning low-level acoustic properties, positional information, and abstract linguistic features up to the level of semantics. It should be noted however that latent indexes exhibit complex behavior with both splitting and absorption. A latent associated with a feature may not be activated for all samples containing that feature. There may be other latents with more narrow feature characteristics. See ([Chanin et al., 2025](#)) for more detail description of feature absorption and splitting.

We present our results organized by feature type: language features, phonetic features, lexical features, syntactic and morphological features, and finally semantic features. The focus is entirely on linguistic information, however there is evidence that non-linguistic information is contained within the latent space, see [Appendix C](#).

4.1 Language Feature

The effect of language was examined by looking at the differences in the latent spaces between Spanish and English samples in the dataset. A simple analysis was performed to evaluate whether or not the presence of single activation of a *latent index* in any frame of an utterance could discriminate between English and non-English utterances. This approach led to the discovery of *latent index 5106*, which allows the classification of non-English samples with a precision of 74.7% with a recall of 91.2%. Limited further analysis reveals that many of the false positives appear to be non-native English speakers.

4.2 Phonetic Features

The Whisper encoder also encodes finer-grained linguistic representations. While it seems necessary for Whisper to represent phonetic information in some manner, it is unclear how exactly it does so. For example, it is unclear whether individual phone-level representations are sufficient or whether richer, context-specific representations are required (see [Ambridge, 2020](#), for an in-depth discussion).

This is further complicated by co-articulation – the phenomenon in which the acoustics of phones can change depending on their surrounding phones ([Daniloff & Hammarberg, 1973](#)). This suggests that Whisper may represent the acoustic linguistic stimuli in terms of larger-level representations such as diphones or triphones, rather than individual phones. Indeed, after examining the latent space, it is clear that the model learns even more complex representations. The SAE has encoded both phones, diphones, and potentially triphones. [Figure 2](#) demonstrates an example of a latent representing the diphone R UW1 (precision value of 88.7%, recall value of 64.9%). Many further examples can be found in [Appendix B](#).

4.3 Lexical Features

The existence of phonetic features is intuitive given that the system’s goal is to transcribe speech. However, lexical features also seem possible especially in English where there is not a 1-to-1 mapping between orthographic representations and phonetic representations. For example, the letter “a” in (American) English can represent several different vowels, such as /æ/ (as in *cat*), /ɑ/ (as in *father*), /eɪ/ (as in *cake*), /ɛ/ (as in *many*), and ə (as in *about*). As a result, it may be advantageous for Whisper to encode lexical information in addition to phonetic information. Indeed, this seems to be the case, as we see features that do seem to be convincingly lexical such as *latent index 28*, which represents the word *his* and has a precision value of 99.3% and a recall value of 81.4%.

4.4 Syntactic and Morphological Features

Linguistic features such as representations of morphological elements are not strictly necessary to transcribe audio, however, given that the sparse autoencoder seems to encode individual words, the model may also encode other abstract representations. An examination of the SAE confirms that the model does seem to be learning morphological representations. [Figure 3](#) demonstrates the words which have the largest ratio of activated frames for *latent index 29*. This latent index correlates with the suffix ‘ly’, which in English serves as an adverbial marker (precision value of 87.2%, recall value of 17.8%). While the recall value is somewhat low for this latent, this is likely due to feature splitting: there are multiple different ‘ly’ adverbial latents, such as latent 9779 (See Appendix for an example B).

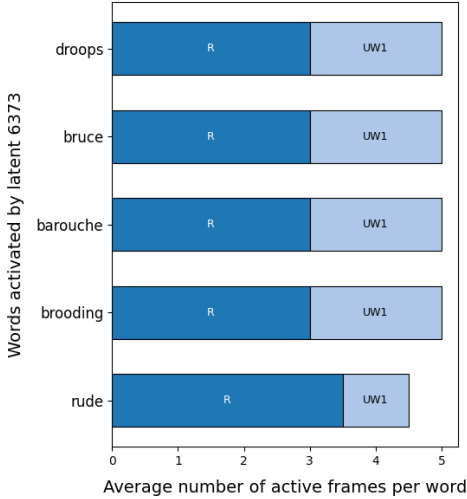


Figure 2: The top 5 words with the highest average number of frames with *latent index 6373* activated (precision value of 88.7%, recall value of 64.9%). The different colored bars represent the average amount of activated frames that each phone in the word received. Phones absent in plot were not activated for this latent.

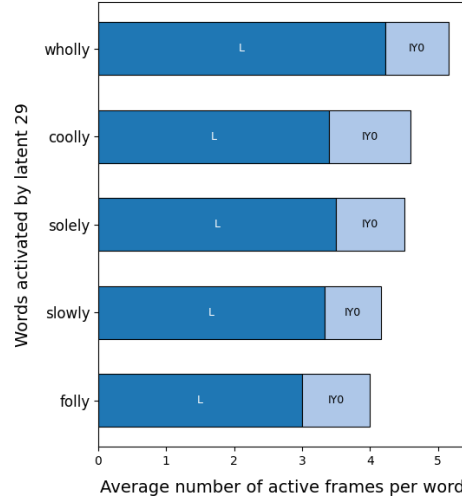


Figure 3: The top 5 words with the highest average number of frames with *latent index 29* activated (precision value of 87.2%, recall value of 17.8%). The different colored bars represent the average amount of activated frames that each phone in the word received. Phones absent in plot were not activated for this latent.

4.5 Semantic Features

Despite not being strictly necessary for transcription, semantic representations may help resolve ambiguities such as homophones. Even for words without homophones, excessive noise or other interference could result in poor transcription which can sometimes be

resolved with appropriate contextual understanding. Notably, the SAE reveals that such semantic representations are indeed present in the Whisper encoder. Two types of semantic features are explored: numbers and profanity.

4.5.1 Numbers

One prevalent type of semantic feature in the dataset is numerical values. Numbers in audio samples were tagged and then examined using the Whisper SAE. In Table 2, activation of *Latent index 7710* is shown which has a strong association with many different numbers with a precision of 38.1% and a recall of 79.2%. It should be noted that of the 7,881 false positives, 3,069 of these occur in utterances which contain numbers. It seems likely that semantic information is not very well temporally bounded within the Whisper encoding. For example, in "one of ten years' transportation", the word 'of' also activates the Numbers feature. Furthermore, the recall of this feature is impacted by feature splitting: out of 1272 false negatives, 664 (52.2%) are 'one' and 'first', 315 (24.8%) are 'two' and 'second', presumably these most frequent numbers have their own latent features.

and the other eighteen by fifteen and they nominally held twenty-two persons
The state side contained twelve good-sized rooms
The frequency and extent of processes against debtors seventy or eighty years ago will appear almost incredible
It generally ran to about six pounds per week
and there were as many as three hundred and fifty prisoners in at one time
This was the chapel yard with its five wards which were calculated to hold seventy prisoners but often held more

Table 2: Activation of *latent index 7710* for samples containing numbers. The darker highlighting shows a stronger activation value.

4.5.2 Profanity

Curse words are another semantic group which can be used in a wide variety of ways and as special intensifiers. Similarly, the set of 331 utterances containing profanity was curated from Mozilla Common Voice English. Through examining the encodings of these audio samples, it appeared that *latent index 3584* was an effective identifier of the presence of swear words, with a recall of 35.4%. Further examination revealed that a significant number of utterances containing the word 'damn' were better represented using a separate latent, *latent index 104*. Table 3 shows a sample of utterances which have activations of *latent index 3584* and *latent index 104*. Together these latents account for a recall of 89.7% with a precision of 6.6%. It should be noted that profanity is quite rare in this dataset, occurring in 0.11% of the entire corpus, based on a word-level count. The false positive rate is only 1.45%. It was also noticed that many of the false positives occur in utterances which contain profanity, again suggesting that the activation of these semantic latents may not be strongly temporally constrained, with some utterances having activations through many frames of the encoding. It is also possible that these latents may more generally represent anger or intensity, often co-occurring with swearing. For example, one utterance with the transcript "With clenched fist, he hit the table." has an activation of one of these latents, despite an absence of swear words.

Alice who the fuck is Bob
Yes fuck
That damn mouse could sense my anger
fuck Ricardo it's just that
Tom can't keep his shit together

Table 3: Activation of *latent index 3584* and *latent index 104* for samples containing profanity.

To confirm that *latent index 3584* is causally linked to profanity rather than merely correlated with it, we employed activation steering. By taking the activation of *latent index 3584* and setting it to a large negative value where it was previously positive, the curse word can be replaced with a phonetically similar word. This modification can also occasionally change following words, since the Whisper decoder follows the most likely subsequent token. Furthermore, by activating *latent index 3584*, setting the 0 value to a large positive number,

curse words can be inserted. In practice, all of the frames of the encoding have had *latent index 3584* activated, leading to the occasional insertion of curse words. Table 4 shows a few examples of steering in either direction. As with the de-activation of profanity, swear words seem to appear more often in phonetically similar words.

Original	Steered
<i>Deactivating Profanity Feature</i>	
Alice, who the fuck is Bob?	Alice, who the focus bar?
Yes, fuck .	Yes, Falk.
That damn mouse could sense my anger.	That dam mouse could sense my anger.
Fuck Ricardo, it's just that.	If we have a Ricardo, it's just that.
Tom can't keep his shit together.	Tom can't keep his sheet together.
<i>Activating Profanity Feature</i>	
I want to listen to the track close to the edge.	I want to listen to the track close to the shit .
You need the whole story.	You need the hell of story.
We can't leave Greg here.	We can't leave, crap , care.
The quick fox jumped on the sleeping cat.	The quick fucks jump on the sleeping cat.
The harder you work, the luckier you are.	The hell you work the fuck you are.

Table 4: Effect of steering *latent index 3584* to deactivate and activate profanity.

4.5.3 Cross-Lingual Activation

The results of Templeton et al. (2024) suggest that many latent indices may have semantic activations that are independent of language. Since Whisper is trained to be a multilingual ASR, it is possible to examine the same effect by curating audios with curse words in Spanish and French using Mozilla Common Voice. Table 5 demonstrates that *latent index 3584* is similarly capable of steering in all three languages, here shown avoiding swear words in the resultant transcript from the steered latent vector. This cross-lingual generalization is particularly notable for French, which was entirely absent from the SAE training data, suggesting that *latent index 3584* encodes language-independent semantic information inherent to the Whisper encoder.

5 Discussions

The present results demonstrate the effectiveness of the SAE as a tool to explore the representations of the Whisper ASR model. More specifically, the present study reveals that the Whisper encoder represents a surprisingly rich hierarchy of information — spanning language, phonetics, lexical items, morphology, and semantics. The emergence of such abstract linguistic structure in a model trained purely on transcription suggests that rich internal representations are a natural consequence of learning to map speech to text at scale. A broad examination of the latent space reflects this; Appendix B shows a subset of the features discovered.⁵

In principle, Whisper only needs phonetic information to accurately transcribe the speech, but there are several advantages to encoding additional information. Many different sentences can have near identical phonetics. For example, *the sky is cool* and *this guy is cool* are quite similar phonetically which can provide a challenge for transcription. However, if the context of the previous sentence was about the warmth of the sun, then it is clear what the correct interpretation is. In other words, semantic information provides an additional tool to disambiguate sentences with multiple potential interpretations.

The encoding of multiple information sources may also reflect the inherent noisiness of speech. Conversations are frequently interrupted by environmental noise, competing speakers, or degraded signal quality, yet both humans and ASR systems must remain robust to such interference. Noisy-channel models of human language processing propose that listeners overcome this by actively combining multiple cues (phonetic, lexical, semantic, and contextual) to recover the most likely intended message from a degraded signal (Gibson

⁵SAE parameters such as latent dimension will affect the granularity of features recovered; for example, a smaller dimension would likely yield more monophonic representations.

Original	Steered
<i>French</i>	
littéralement Darwin est une salope 	L'étréale-midi L'étréale-midi
Ou ça ? Dans ton cul ?	ou ça, dans tant que du tout
C'est vraiment un gros con , on l'y en a	C'est vraiment un grand congrès
Mais ce qui va chier , dehors	Mets ce qui va chez deux ors
Tu vas voir ta gueule !	Tu vas voir ta gagne
<i>Spanish</i>	
Lo he oído, pendejo	Lo he oído en deajo
Con una mano me basta para patiarle el culo	Con una mano me basta para apatirte el color
Entras en este ciclo de zorra	entre en este ciclo de sorra
¡Cómo mucho! ¡Dios, Joder!	Como mucho. Díos, Jadera
más seguro es en la cabina, será idiota	Más seguro en la cabina, se ha ido a la idea

Table 5: French and Spanish samples steered using *latent index 3584* to ‘deactivate’ profanity.

et al., 2013). It is possible that Whisper, having been trained on naturalistic and noisy audio at scale, has converged on a similar strategy, encoding redundant information across multiple levels of abstraction as a form of noisy-channel robustness.

Finally, the SAE lends itself to important debates within the linguistics literature. One such debate is about the degree of memorization in language models as opposed to abstraction (Houghton, 2025; Houghton et al., 2025; Jian & Manning, 2026). The SAE demonstrates that abstract representations such as those for profanity and numbers are represented in Whisper. Further, the classification of French profanity into the profanity latent shows that this abstraction is not an artifact of the SAE since the SAE was not trained on French data. This suggests that at least some abstract information is encoded in a language-agnostic manner within the Whisper encoder itself, rather than emerging from the SAE or the training data.

6 Conclusion and Future Work

This study demonstrates the ability of Sparse Autoencoders as tools to interrogate the inner workings of Whisper, an end-to-end ASR model. Basic building block features such as phonetic and positional information were identified. However, increasingly complex and abstract concepts like morphemes, words and even semantic information were discovered to be present.

Future work may explore whether these same features are present among other end-to-end ASR models or even between different sizes of the Whisper architecture. Additionally, explorations of different latent dimension sizes may hint at the atomic concepts which these models encode. Finally, training SAEs at different layers within ASR models may also reveal how different types of information are propagated throughout ASR models. For example, it seems likely that the type of information represented at each layer of the encoder may vary. SAEs provide an avenue to explore how information develops and evolves throughout the model.

6.1 Limitations

This work has three main limitations, mostly based around resources. First, the work has only focused on a single sparse autoencoder. Future work will focus more on examining SAEs built with differing latent dimensions and understanding how the features of the latent space evolve. Second, only the Whisper base model was investigated, which is one of the smallest Whisper models. It would be interesting to understand if larger Whisper models encode information differently, or in fact if other end-to-end ASR models contain the same depth of semantic information. Finally, although some non-English data was used, English was the primary focus of the study.

References

- Ben Ambridge. Against stored abstractions: A radical exemplar model of language acquisition. *First Language*, 40(5-6):509–559, 2020.
- Rosana Ardila, Megan Branson, Kelly Davis, Michael Kohler, Josh Meyer, Michael Henretty, Reuben Morais, Lindsay Saunders, Francis Tyers, and Gregor Weber. Common voice: A massively-multilingual speech corpus. In Nicoletta Calzolari, Frédéric B  chet, Philippe Blache, Khalid Choukri, Christopher Cieri, Thierry Declerck, Sara Goggi, Hitoshi Isahara, Bente Maegaard, Joseph Mariani, H  l  ne Mazo, Asuncion Moreno, Jan Odijk, and Stelios Piperidis (eds.), *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pp. 4218–4222, Marseille, France, May 2020. European Language Resources Association. ISBN 979-10-95546-34-4. URL <https://aclanthology.org/2020.lrec-1.520/>.
- Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, et al. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning*, pp. 2397–2430. PMLR, 2023.
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermy, Tom Conerly, Nicholas L Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Chris Olah. Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread, October 2023a. URL <https://transformer-circuits.pub/2023/monosemantic-features>. Anthropic.
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermy, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023b. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- David Chanin, James Wilken-Smith, Tom     Dulka, Hardik Bhatnagar, and Joseph Isaac Bloom. A is for absorption: Studying feature splitting and absorption in sparse autoencoders, 2025. URL <https://openreview.net/forum?id=LC2KxRwC3n>.
- Raymond G Daniloff and Robert E Hammarberg. On defining coarticulation. *Journal of Phonetics*, 1(3):239–248, 1973.
- Leo Gao, Tom Dupre la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=tcsZt9ZKND>.
- Edward Gibson, Leon Bergen, and Steven T Piantadosi. Rational integration of noisy evidence and prior semantic expectations in sentence interpretation. *Proceedings of the National Academy of Sciences*, 110(20):8051–8056, 2013.
- Yuan Gong, Sameer Khurana, Leonid Karlinsky, and James Glass. Whisper-at: Noise-robust automatic speech recognizers are also strong general audio event taggers. In *Interspeech 2023*, pp. 2798–2802, 2023. doi: 10.21437/Interspeech.2023-2193.
- Adriana Guevara-Rukoz, Isin Demirsahin, Fei He, Shan-Hui Cathy Chu, Supheakmungkol Sarin, Knot Pipatsrisawat, Alexander Gutkin, Alena Butryna, and Oddur Kjartansson. Crowdsourcing Latin American Spanish for Low-Resource Text-to-Speech. In *Proceedings of The 12th Language Resources and Evaluation Conference (LREC)*, pp. 6504–6513, Marseille, France, May 2020. European Language Resources Association (ELRA). ISBN 979-10-95546-34-4. URL <https://www.aclweb.org/anthology/2020.lrec-1.801>.

-
- Coleman Haley. This is a bert. now there are several of them. can they generalize to novel words? In *Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pp. 333–341, 2020.
- Carlos D. Hernandez-Mena. TEDx Spanish Corpus. Audio and transcripts in Spanish taken from the TEDx Talks; shared under the CC BY-NC-ND 4.0 license. Web Download, 2019.
- Zachary Nicholas Houghton. *Multi-Word Representations in Minds and Models: Investigating the Storage of Multi-Word Phrases in Humans and Large Language Models*. PhD thesis, University of California, Davis, 2025.
- Zachary Nicholas Houghton, Kenji Sagae, and Emily Morgan. The role of abstract representations and observed preferences in the ordering of binomials in large language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 695–702, 2025.
- Robert Huben, Hoagy Cunningham, Logan Riggs Smith, Aidan Ewart, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=F76bwRSLeK>.
- Keith Ito and Linda Johnson. The lj speech dataset. <https://keithito.com/LJ-Speech-Dataset/>, 2017.
- Jasper Jian and Christopher D Manning. Humans and transformer lms: Abstraction drives language learning. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 752–765, 2026.
- Karim Lasri, Olga Semnck, Alessandro Lenci, and Thierry Poibeau. Subject verb agreement error patterns in meaningless sentences: Humans vs. bert. *arXiv preprint arXiv:2209.10538*, 2022.
- Bingzhi Li and Guillaume Wisniewski. Are neural networks extracting linguistic properties or memorizing training data? an observation with a multilingual probe for predicting tense. In *EACL 2021*, 2021.
- Alireza Makhzani and Brendan Frey. k-sparse autoencoders, 2014. URL <https://arxiv.org/abs/1312.5663>.
- Kanishka Misra and Kyle Mahowald. Language models learn rare phenomena from less rare phenomena: The case of the missing aanns. *arXiv preprint arXiv:2403.19827*, 2024.
- J. Morgan. West point heroico spanish speech LDC2006S37. Linguistic Data Consortium, 2006. URL <https://openslr.org/39/>. [Online].
- A. Nagrani, J. S. Chung, and A. Zisserman. Voxceleb: a large-scale speaker identification dataset. In *INTERSPEECH*, 2017.
- Dingyi Pan and Ben Bergen. Are explicit belief representations necessary? a comparison between large language models and bayesian probabilistic models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 11483–11498, 2025.
- Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. Librispeech: An asr corpus based on public domain audio books. In *2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 5206–5210, 2015. doi: 10.1109/ICASSP.2015.7178964.
- Karol J. Piczak. Esc: Dataset for environmental sound classification. In *Proceedings of the 23rd ACM International Conference on Multimedia, MM ’15*, pp. 1015–1018, New York, NY, USA, 2015. Association for Computing Machinery. ISBN 9781450334594. doi: 10.1145/2733373.2806390. URL <https://doi.org/10.1145/2733373.2806390>.

-
- Daniel Pluth, Yu Zhou, and Vijay K. Gurbani. Sparse autoencoder insights on voice embeddings. In *2025 Conference on Artificial Intelligence x Multimedia (AIxMM)*, pp. 1–6, 2025. doi: 10.1109/AIxMM62960.2025.00007.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision, 2022. URL <https://arxiv.org/abs/2212.04356>.
- David Snyder, Guoguo Chen, and Daniel Povey. MUSAN: A Music, Speech, and Noise Corpus, 2015. arXiv:1510.08484v1.
- Xiaoqing Sun, Alessandro Stolfo, Joshua Engels, Ben Wu, Senthoooran Rajamanoharan, Mrinmaya Sachan, and Max Tegmark. Dense sae latents are features, not bugs, 2025. URL <https://arxiv.org/abs/2506.15679>.
- Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermyn, Shan Carter, Chris Olah, and Tom Henighan. Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer Circuits Thread*, 2024. URL <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>.
- Ian Tenney, Dipanjan Das, and Ellie Pavlick. Bert rediscovers the classical nlp pipeline. *arXiv preprint arXiv:1905.05950*, 2019.
- Qing Yao, Kanishka Misra, Leonie Weissweiler, and Kyle Mahowald. Both direct and indirect evidence contribute to dative alternation preferences in language models. *arXiv preprint arXiv:2503.20850*, 2025.

A Domain Space

Let $E : \mathcal{X} \rightarrow \mathcal{Z}$ denote a fixed Whisper encoder, mapping inputs from a high-dimensional space $\mathcal{X}$ to a latent space $\mathcal{Z}$. The encoder’s weights define an image manifold $\mathcal{M} = \{E(x) : x \in \mathcal{X}\} \subset \mathcal{Z}$, a structured subset of $\mathcal{Z}$ whose geometry is entirely determined by E . A sparse autoencoder trained on the outputs of E operates exclusively in $\mathcal{Z}$ -space and therefore has access only to sampled points on $\mathcal{M}$. Critically, $\mathcal{M}$ is invariant to any choice of SAE training data: an ablation that removes a semantic subset of $\mathcal{X}$ modifies the sampling measure over $\mathcal{M}$ but cannot alter $\mathcal{M}$ itself. The SAE cannot recover structure that E does not encode, nor can it fail to recover structure that E does encode due to ablation alone, provided coverage of the manifold is sufficient.

An analogous invariance holds for SAE hyperparameters. The latent space dimensionality and sparsity constraint parameterize a family of approximations $\varphi_\theta(\mathcal{M})$ to the manifold, but varying θ does not alter $\mathcal{M}$. Increasing latent dimensionality improves the granularity of the decomposition; relaxing sparsity allows greater feature entanglement per activation. Both affect approximation fidelity, not the underlying representational geometry. Consequently, ablation studies over training data composition and hyperparameter searches are properly understood as studies of approximation error under a fixed target $\mathcal{M}$, rather than as explorations of what structure is in principle recoverable from E . The object of interest, the geometry of $\mathcal{M}$ as determined by the encoder (representing the discovered features), is not a variable in either type of experiment.

B Latent Samples

latent_id	label	correct
147	sentence-initial word (first word of each sentence)	1
570	voiceless alveolar fricative /s/ (spelled s, c, x, ss, etc.)	1
604	detects unstressed vowel (schwa-like /ə/)	1
668	presence of the /ɜ:/ (schwa-r) vowel sound (as in "er/ir/ur" syllables)	0
722	words containing the digraph "sh" or "ch"	0
730	words containing the /dʒ/ (j/soft-g) sound	1
994	words containing the /k/ (hard c/k/ck) sound	1
1052	back-rounded vowel /ɔ/ (open-mid back rounded vowel)	1
1288	words with the long-A /eɪ/ vowel (face/race rhyme family)	0
1579	detects the word "hundred" (including plural and compound forms)	1
1626	presence of a nasal consonant (/n/ or /m/)	1
1799	detects the orthographic bigram "so"	1
1822	presence of final /s/ sound	0
1992	detects the Latin root "clud/clus" (include, exclude, conclude, etc.)	1
2127	voiceless velar stop /k/ phoneme	1
2151	determiner (articles, demonstratives, numerals)	1
2219	detects words containing the letter "f"	1
2318	detects the lateral consonant /l/ (letter l)	1
2347	detects the /s/ fricative (s-sound)	1
2353	presence of the long-A diphthong /eɪ/ (as in "state", "great", "candidate", "tolerate")	0
2561	presence of the velar nasal /ŋ/ (ng/nk sound)	1
2875	initial stop-liquid consonant clusters (e.g., /pr/, /br/, /fr/, /dr/)	0
3202	words containing the /s/ sound (voiceless alveolar fricative)	1
3559	words ending in "s" (plurals/possessives)	1
3652	dental fricative /θ/ (th) sound	0
3714	detects -ed past-tense/participle suffix	1
3733	words containing a consonant cluster (two adjacent consonants)	0
3920	activation on short function words and morphemes (auxiliaries, prepositions, articles)	0
4413	activation on words containing the alveolar stop /t/ (or /d/)	0
4555	detects the /n/ nasal sound	1
4808	first vowel (early vowel sound) of a word	0
4948	short-i vowel /ɪ/ (high front lax vowel)	0
5021	detects back-rounded vowel /o/ (e.g., 'o'/'ow' sound)	1
5045	presence of the /t/ phoneme (letter t)	1
5060	detects the morpheme "with-" (with, within, without, forthwith)	1
5134	presence of a sibilant fricative (/s/, /z/, /ʒ/)	1
5237	detects the prefix "after-" (e.g., after, afterwards, thereafter, afterwork, afternoons)	1
5268	detects rhotic /r/ sound (letter r)	1
5349	presence of the /s/ sound (voiceless alveolar fricative)	1
5352	presence of /s/ or /z/ (alveolar fricative) sound	1
5629	detects the /st/ consonant cluster	1
5633	detects the /v/ (voiced labiodental fricative) sound	1
6079	detects common Latin-derived affixes (com-, de-, re-, dis-, su-, -es)	0
6136	detects the -ing participial/gerund suffix	1
6206	first word after punctuation (clause-initial)	0
6334	detects the /tʃ/ (ch) sound	1
6411	negation particles (not / no)	1
6413	presence of bilabial nasal /m/ sound	1
6417	presence of the schwa vowel /ə/ in the word	0
6692	function words – articles, prepositions, demonstratives, pronouns	1
6775	rhotic consonant /r/ (r) activation	1
7289	detects the /r/ consonant	1
7507	detects the lateral /l/ sound	1
7781	detects the /v/ (voiced labiodental fricative) sound	1
7867	short function words (e.g., "of", "a", "by", "off")	1

Continued on next page

latent_id	label	correct
7954	detects the morpheme “hundred” (numeric multiplier)	1
8229	presence of a nasal consonant (e.g., /n/ or /m/) in the word	1
8310	detects the /ru:/ (long-u) sound	1
8365	presence of the lateral /l/ sound (letter L)	1
8433	words containing a voiced labial consonant (/b/ or /v/)	1
8555	third-person plural pronoun “them” and its inflections	1
8748	definite article “the” (including capitalised forms)	1
8765	words containing the /saɪ/ (s-ai) sound	1
8782	closed-class function words (prepositions, conjunctions, determiners)	0
8785	affricate consonant /tʃ/ or /dʒ/ (ch/j)	1
9293	detects the /num/ (or /nom/) sound cluster	1
9722	presence of the high front vowel /i/ (ee/eye sound)	1
9779	detects the -ly suffix (adverbial/derived -ly words)	1
9794	presence of the /s/ (alveolar fricative) phoneme	1
10429	detects presence of the /n/ nasal sound (letter n)	1
10523	definite article “the” (and similar determiners)	1
10668	punctuation (comma, semicolon, dash) and following word	0
10724	detects the word “time” (including plural, capitalized, and compounds)	1
11705	detects the rhotic /r/ sound	1
12424	words containing the diphthong /aɪ/ (mind, mine, might, line, turbine, etc.)	1
12477	detects rhotic /r/ sound (letter r)	0
12689	presence of the /kl/ consonant cluster	1
12697	sentence-initial word (first token)	1
12844	presence of the letter “o” in a word	0
12963	words containing the /ɔn/-/wɒn/ sound (e.g., one, won, on, wan, want, worn)	1
13064	detects the /m/ (in) sound or grapheme	1
13432	words ending in a nasal consonant (/m/, /n/, /ŋ/)	1
13696	words containing the postalveolar affricate /tʃ/ or /dʒ/ sound	1
13724	initial consonant clusters (st, tr, cr, pr, sh, cl, gr, etc.)	0
14001	words with the /eɪk/ sound (make, take and inflections)	1
14039	detects the word “and” (coordinating conjunction)	1
14157	closed-class function words (prepositions, articles, auxiliaries, conjunctions)	0
14373	detects the /wel/ phoneme cluster (e.g., “well”, “twelve”, “Welsh”, etc.)	1
14601	words containing the alveolar fricative /s/ (or its voiced counterpart /z/)	1
14752	words containing a voiced sibilant (/z/ or /ʒ/)	1
14770	detects bilabial stop consonants (/b/ or /p/)	1
14851	initial vowel-onset words	0
14869	first-person possessive determiner “my” (including “myself”)	1
15083	rhotic /ɔ:r/ vowel (or/ore/oor) sound	1
15123	voiced labiodental fricative /v/	1
15387	presence of a labial consonant (b / p / v)	1
15606	onset of primary stress syllable (initial consonant or cluster)	0
15749	short function words (prepositions, conjunctions, particles)	1
15915	presence of the /ʃ/ (sh) sound	1
15939	presence of alveolar liquid consonants (/l/ or /r/)	0

C Non-Linguistic Latents

C.1 Noise

As previously mentioned, noise information has been shown to be present in the Whisper encoded data, however it was not clear how this information would be encoded within the SAE latent space as an activation. Noisy samples were sourced from ESC-50 which contains 50 distinct types of non-speech noise (Piczak, 2015). By comparing voiced and non-voiced samples, the activation of *latent index 15019* clearly discriminated the two classes. A predictor was created according to the following logic:

$$pred = \begin{cases} \text{if } \ell_{15019} > 0 & \text{noise} \\ \text{else} & \text{voice} \end{cases}$$

Figure 4 shows the confusion matrix created using this predictor to compare to the ground truth. This predictor has a 97.7% precision for the noise audio samples.

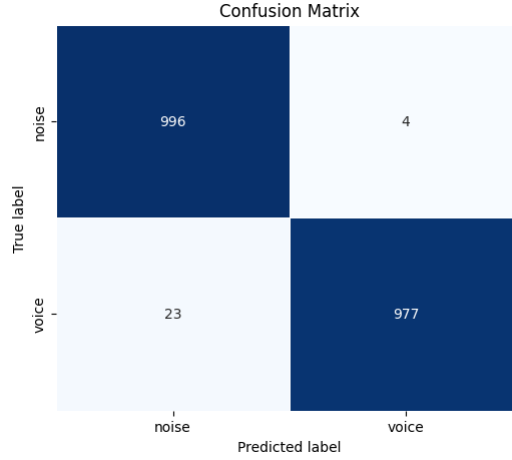


Figure 4: *Latent index 15019* is associated with noise only audio, the latent index can effectively separate noise-only audio from speech audio.

C.2 Positional Features

The Whisper encoder uses a positional encoding which is added to the input features, therefore some positional information likely exists in the encoding. However, since the SAE encodes all information as discrete latent indexes, it is not clear how the positional information would manifest. Figure 5 shows three examples of positional latent indexes, all of which are tightly located temporally. *Latent index 12816* has an mean activation timestamp of 8.44 seconds and a standard deviation of 0.19. *Latent index 15039* is at 10.09 seconds on average with an SD of 0.04, and *latent index 3604* activates at 14.04 seconds with an SD of 0.37. This positional representation is potentially useful for the decoder to determine word order since the encoding is passed to the decoder with cross-attention.

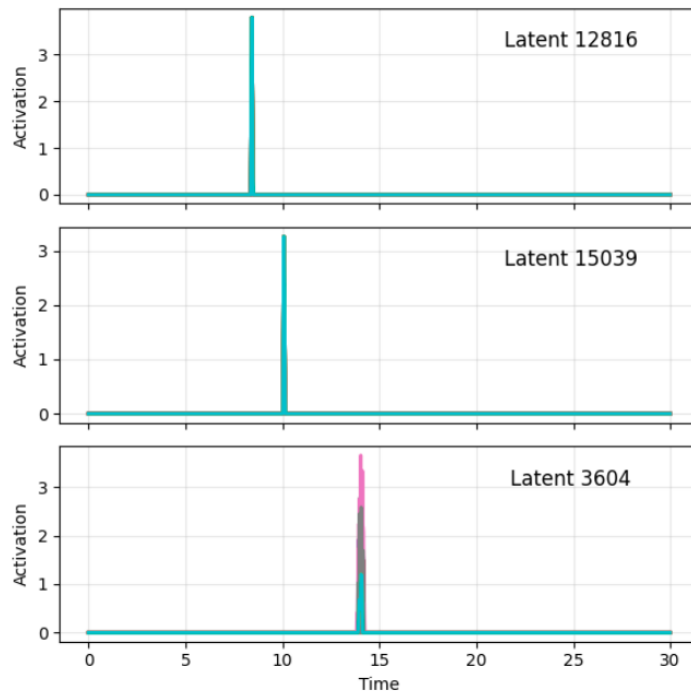


Figure 5: Three different positional latents showing activations on 20 different audio files. These positional latents are very temporally constrained and consistent between distinct files.